Braneworld Shortcuts with paceval: Geometric Null Paths, Five-Dimensional Wave Fields, and Cosmological Travel Times

Integrated numerical research study – methods, independently checked results, and scientific limitations

paceval research – computational follow-up investigation

6 November 2026

Executive Summary

The earlier **paceval study of theoretical traversable wormholes** [11] explored the possibility of connecting distant regions of space through a wormhole throat. It generated and compared mathematical geometries using objectives related to null-energy-condition (NEC) violation and spacetime curvature. The present work adopts a different framework: **an expanding four-dimensional brane embedded in a five-dimensional anti-de Sitter (AdS_5) spacetime**. Rather than constructing a wormhole, it investigates a path that leaves the brane, travels through the higher-dimensional bulk, and rejoins the brane at the receiver.

Main geometric result. In one selected, dimensionless high-energy model, the bulk null route has a **20.96279244% shorter receiver travel time** than light confined to the brane. This means that, *under the model assumptions*, a hypothetical massless signal capable of following that bulk trajectory would reach the same comoving receiver approximately one fifth earlier. **This percentage is neither a universal lower bound nor a prediction for the present-day Universe.** The 20.96% is measured relative to the brane-confined travel time; it should not be described as a 20.96% increase in local signal speed.

Neither route exceeds the local speed of light. The difference comes from their permitted paths through the full geometry. **Faster-than-light messaging with ordinary light has not been demonstrated.** Standard-Model photons are generally confined to the brane in the class of models considered. The physical excitation of a suitable bulk mode, its propagation, and its detectable return to the brane remain open issues.

The conceptual advantage over a classical traversable wormhole is that **no traversable throat needs to be constructed, and the brane matter in the selected example does not need to violate the NEC in the manner required at a Morris–Thorne wormhole throat**. For positive matter density and $w = -0.5$, the perfect fluid satisfies $ ho+p= ho(1+w)>0$. The model nevertheless assumes a five-dimensional AdS geometry, a negative bulk cosmological constant, a fine-tuned positive brane tension, and particular boundary and matching conditions. Therefore, **“no exotic throat matter” is not a claim of general physical feasibility or an engineering energy budget**.

Additional strengths include a direct emitter–receiver comparison, a comparatively well-defined background geometry, connections to cosmological expansion, and a route toward investigating five-dimensional gravitational-wave fields. The numerical **C++ programs** use reusable paceval execution graphs and were checked against separate numerical reference methods.

Three connected questions guide the study: **(1)** Do geometrical bulk shortcuts occur in the specified model? **(2)** Can five-dimensional wave-field evolution with a moving brane be computed and cross-checked? **(3)** How do purely geometrical delays behave under illustrative cosmological scaling? In a separate 40-Mpc model scenario, the geometrical delays are of order seconds. **A grid-independent demonstration of an earlier arriving, physically observable gravitational-wave signal has not been obtained.**

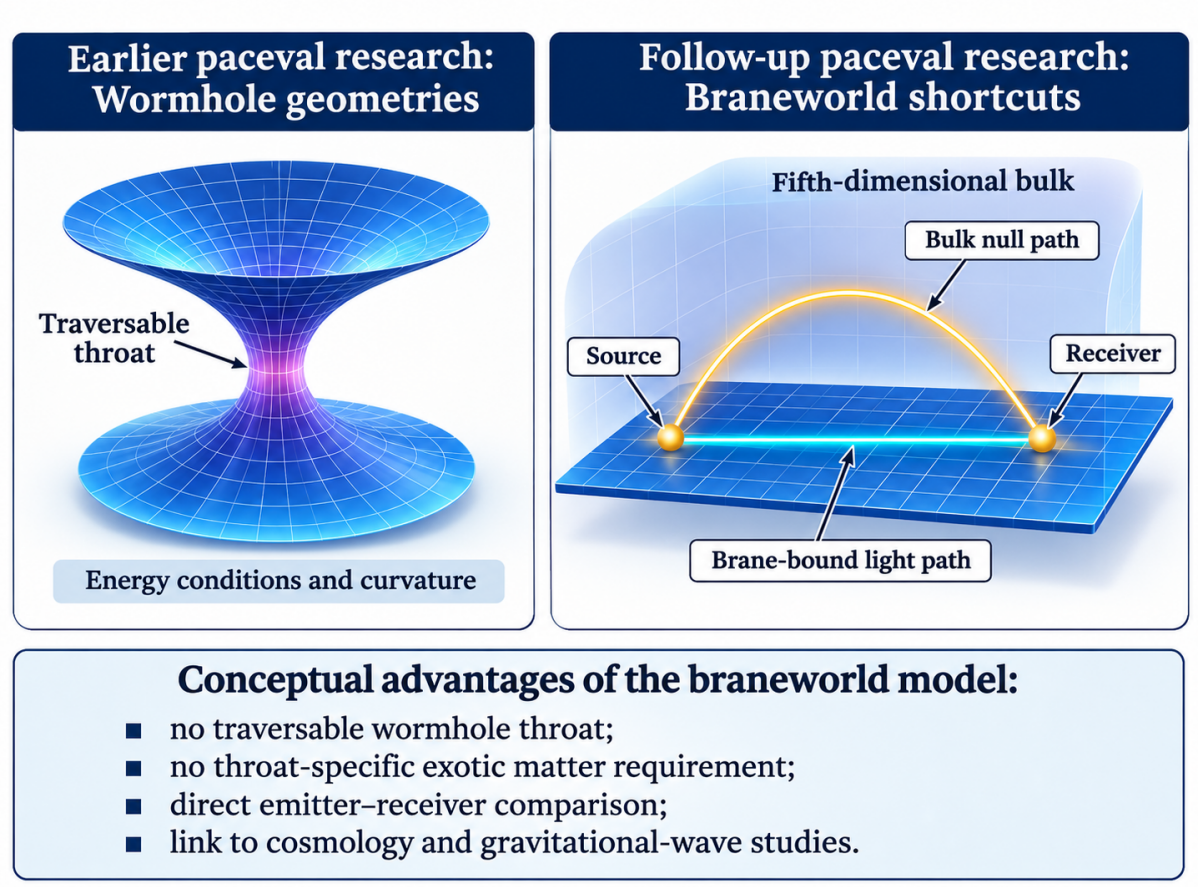


Figure 1: Comparison of the earlier wormhole study with the braneworld investigation. Conceptual 3D illustration only; not a scaled representation of five-dimensional spacetime.

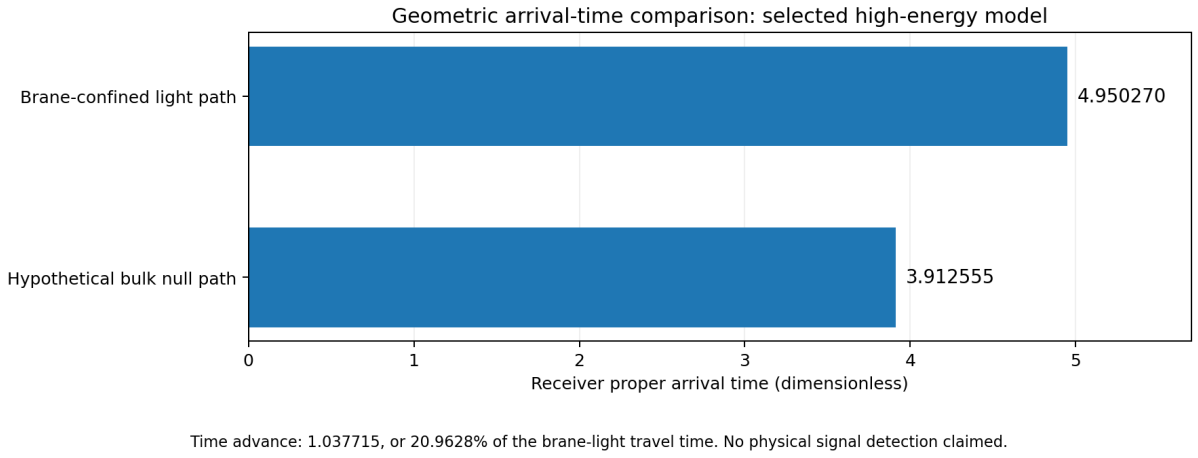


Figure 2: Receiver proper travel times for the two mathematical routes in the selected high-energy model.

The strongest outcomes are therefore a **reproducible conditional geometric shortcut**, a **numerically verified restricted wave-operator experiment**, and a **high-precision method for small cosmological geometric delays**. The work is a computational basis for further physical investigation, not the discovery of a real superluminal signal.

Abstract

Objective. Investigate possible arrival-time differences between brane-confined light rays and five-dimensional bulk null rays, and determine what a restricted dynamical wave-field calculation and physically scaled geometry do—and do not—establish.

Model and methods. The geometric background is pure AdS$_5$ with a moving FLRW brane, fine-tuned brane tension, and an idealized conserved perfect fluid. A restricted linearized tensor wave operator is evolved numerically with a moving boundary, including Fourier modes and a pulse localized along one brane direction. To examine physical scales, a positive-definite double-integral expression for the geometric range excess is combined with illustrative cosmological expansion. The C++ programs rely on the standard library and paceval, with reusable execution graphs and batch evaluation; SciPy and MPMath are used only for external checks.

Reproduced results. For $ ho_0=10$, $w = -0.5$, and $D = 1$, the dimensionless bulk-null arrival advance is 1.03771487669294, or 20.96279244046836% of the brane-light travel time. An analytic static-wave reference yields maximum errors of 1.59×10^{-9}, 9.91×10^{-11}, and 6.22×10^{-12} under successive time-step refinements. An independently implemented moving-brane pulse calculation differs by at most 2.11×10^{-9} in the selected high-energy comparison. A planar geometric characteristic gives $T_{\text{Bulk}} = 1.25843305$ versus $T_{\text{Brane}} = 1.31751327$ for the nearest source edge; however, the numerical wavefront is **not** established independently of grid resolution and amplitude threshold. In an illustrative 40-Mpc scenario, geometrical advances are 1.59517404892358 s at $\ell = 0.1$ Mpc and 45.65786889658452 s at $\ell = 0.535$ Mpc. Selected Richardson-extrapolated paceval computations agree with independently computed high-precision values at the level of a few trillionths of a second.

Scope. These are preliminary computational findings, not independently peer-reviewed, with idealized background and source assumptions. They do not establish a physical bulk messenger, a measurable early gravitational-wave arrival, an observational upper bound on the AdS length derived by this software, or a faster-than-light communication channel.

1. Research Question, Scientific Context, and Advantages of the New Approach

The Braneworld investigation is a methodological follow-up to the earlier **paceval research on theoretical traversable wormhole geometries** [11], but it examines a fundamentally different geometric construction. The wormhole study searched for static shape and redshift functions under energy-condition and curvature criteria. The present study instead asks whether a higher-dimensional spacetime permits **geometrical shortcuts without a wormhole throat**.

This change is scientifically useful. Classical static traversable Morris–Thorne wormholes require radial NEC violation at the throat under the conventional flare-out conditions. The particular braneworld model studied here constructs no throat; its on-brane perfect fluid has $\rho > 0$ and $\rho + p > 0$ for the benchmark $w = -0.5$. Consequently, the calculation does **not** rely on the classical wormhole-throat exotic-matter mechanism. This does not validate all energy conditions in every sector, prove stability, or eliminate the need for extra-dimensional gravitational assumptions. In particular, the chosen $w = -0.5$ fluid does not satisfy the usual strong energy condition ($\rho + 3p < 0$); the claim is restricted to the NEC and the absence of a wormhole throat.

The selected successful model consists of a moving matter-FLRW brane in **pure AdS$_5$**. Two independently implemented numerical integration approaches and an explicit receiver boundary-value calculation yield, for $\rho_0 = 10$, $w = -0.5$, $D = 1$,

$$\tau_{\text{Brane}} = 4.95027024495861, \qquad \tau_{\text{Bulk}} = 3.91255536826567,$$

$$\Delta\tau = 1.03771487669294, \qquad 100\frac{\Delta\tau}{\tau_{\text{Brane}}} = 20.96279244046836\%.$$

A corrected numerical scan contains **50 evaluable parameter cases**, including 43 with a positive geometric advance for $w > -1$ and seven de Sitter controls with zero advance to the displayed precision. This selected sample is **not** a global optimum, a universal minimum advance, or a realistic cosmological prediction. Its role is to demonstrate reproducibility within the model.

The next steps extend this geometric calculation to **linear five-dimensional wave-field evolution with a moving boundary** and to **cosmologically scaled, very small geometric travel-time differences**. The benefits are a direct source–receiver construction, a well-defined AdS reference background, a link to expansion history, numerical wave-field testing, and careful accounting for astrophysical emission lags.

Bulk shortcuts were discussed in earlier publications by Ishihara (2001), Caldwell and Langlois (2001), and Abdalla and Casali (2002) [1–3]. This work does **not** claim priority for the physical mechanism. Its contribution is a reproducible, paceval-based numerical investigation joining a geometric benchmark to restricted wave-field and cosmological-scale calculations.

Investigated component	Numerical result	Valid scientific interpretation
Moving FLRW brane in pure AdS_5	20.9628% for a selected high-energy example	Conditional geometric advance, not a detected signal
Linear wave fields and planar pulses	Analytic and independent solver checks pass; front not grid-converged	Numerical validation of a restricted wave operator
Cosmological geometric scaling	Delays of order seconds for chosen parameters	Model-dependent geometry, not observational evidence

2. Background Geometry and Linearized Wave Operator

In dimensionless Poincare coordinates of pure AdS5,

$$ds^2 = z^{-2}\left(-dT^2 + dx_1^2 + dx_2^2 + dx_3^2 + dz^2\right), \qquad z > 0.$$

The FLRW brane is a moving boundary at $z = z_b(T) = 1/a(T)$. For a fine-tuned tension and a single conserved perfect-fluid component,

$$H(a)^2 = \frac{\rho(a)}{3} + \frac{\rho(a)^2}{36}, \quad \rho(a) = \rho_0 a^{-3(1+w)}, \quad v(T) = \frac{dz_b}{dT} = -\frac{H}{\sqrt{1+H^2}}.$$

The retained side is $z \geq z_b(T)$ under the specified conditional Randall–Sundrum orientation. The tensor transverse-traceless (TT) bulk equation, after Fourier decomposition along the homogeneous brane spatial directions, is taken as

$$\partial_T^2 h = \partial_z^2 h - \frac{3}{z}\partial_z h - k^2 h. \tag{1}$$

This equation is the free AdS5 *bulk* tensor operator. The full perturbed Israel junction problem can include source anisotropic stress, gauge conditions and brane-bending issues. In the simplified zero-tensor-stress boundary scenario, the normal derivative has the form

$$\partial_z h + v\,\partial_T h = 0 \qquad \text{at } z = z_b(T). \tag{2}$$

The code's optional forcing replaces the right-hand side of (2) by a compact pulse J. **Its physical source normalization and tensor-gauge consistency are not derived**; it is a controlled boundary condition for a mathematical PDE experiment, not a fully specified compact binary merger.

To fix the computational boundary, introduce $y = z - z_b(T) \geq 0$ and define $u(T, y) = h(T, z)$ and $q(T, y) = (\partial_T h)|_z$. Direct application of the chain rule gives

$$\partial_T u = q + v\partial_y u, \qquad \partial_T q = \partial_y^2 u - \frac{3}{y + z_b}\partial_y u - k^2 u + v\partial_y q. \tag{3}$$

The moving boundary is now $y = 0$, where $u_y + vq = J(T)$ is enforced using a ghost cell. The independent Python method-of-lines check uses a separately written finite-difference operator and a SciPy DOP853 temporal integrator. The production C++ source contains **no SciPy/NumPy or equivalent third-party solver**.

For a source localized in one brane coordinate x, the 2D planar experiment instead uses

$$\partial_T q = \partial_y^2 u + \partial_x^2 u - \frac{3}{y + z_b}\partial_y u + v\partial_y q. \tag{4}$$

The modeled source is even in x, with compact support $0 \leq x < R$ and a finite emission interval. In the two remaining brane directions it is taken to be translation invariant. **Equation (4) is therefore not the full 3D spatially localized retarded gravitational-wave problem.** The code uses simple reflecting outer edges; sufficiently early time windows must be selected to avoid contamination from reflected waves.

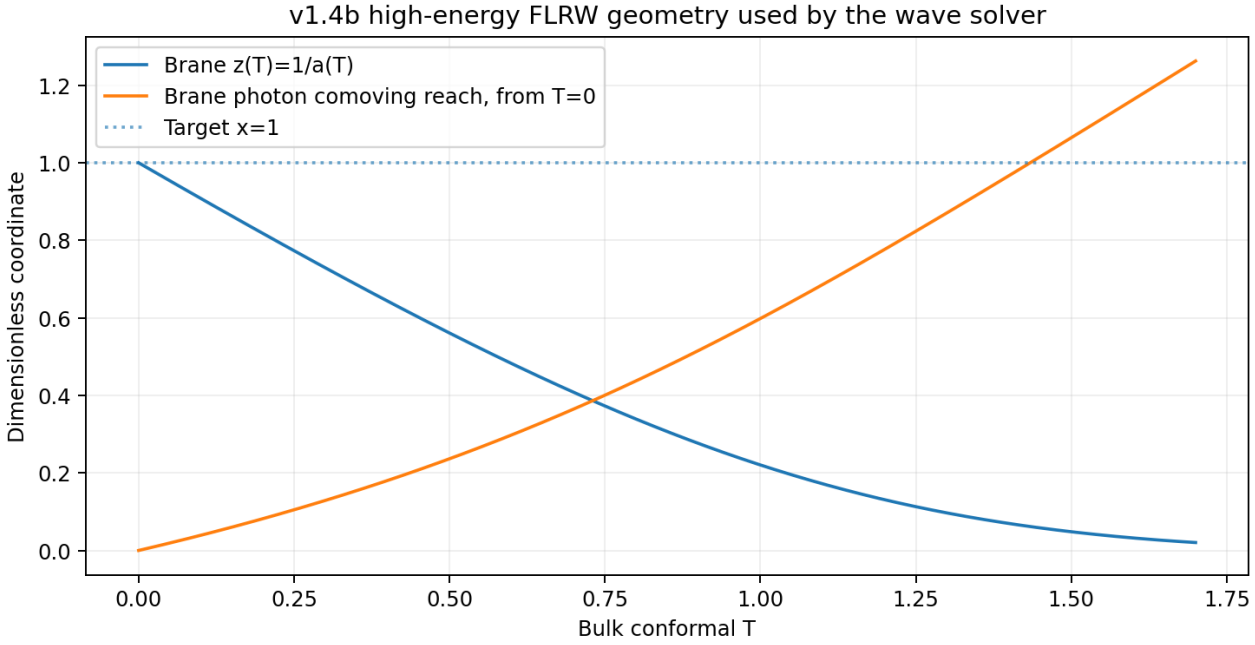


Figure 3: Selected high-energy moving FLRW brane and comoving light-horizon geometry; mathematical background variables, not observed signals.

3. paceval-Based Numerical Implementation

Three separately executable **C++ programs** implement the research calculations:

Program	Scientific task	Role of paceval
Fourier-mode wave solver	Fourier-k mode on the moving one-dimensional y domain	Reused Friedmann graph and two complete PDE right-hand-side graphs evaluated in batches at each RK4 stage
Planar pulse solver	Planar x, y field evolution with finite-duration source	Reused Friedmann graph and two 2D right-hand-side graphs evaluated in large batches
Cosmological geometry solver	Brane geometry on physical scales and illustrative timing comparisons	Reused Friedmann, positive pair-kernel, and chi-square graphs, pairwise batch evaluation and optional Richardson extrapolation

The mathematical expressions are evaluated using reusable paceval execution graphs. Time stepping, moving-boundary handling, grid indexing, and output are implemented in C++. **All wave-operator right-hand-side expressions are evaluated by paceval graphs.** Independent numerical reference calculations are performed outside the production C++ code.

4. Numerical Validation of Wave-Field Evolution

4.1 Exact static-brane reference

For a static brane $v = 0$ and fixed $z_b = 1$, the spatially constant mode $h(T, z) = \cos(kT)$ satisfies (1) and the homogeneous Neumann boundary condition. Using $k = 3$, the maximum absolute field errors over $0 \le T \le 1$ were:

Requested ΔT	Actual ΔT (approximately)	Max absolute error
0.006	0.005988	1.5908×10^{-9}
0.003	0.002994	9.9063×10^{-11}
0.0015	0.001499	6.2175×10^{-12}

Error ratios near 16 support the expected fourth-order time integration for this smooth control, **not** global fourth-order accuracy of the complete spatially discretized driven PDE. For $k = 0$ and a moving brane, a constant field is also an exact solution, and the code reproduced it without printed drift at 81, 161 and 321 spatial cells.

4.2 Independently integrated moving-brane pulses

For $(\rho_0, w) = (0.1, 0)$, spatial mesh 161, a compact time source and $k = 0$, a separately written SciPy method-of-lines implementation agrees with the paceval C++ output at the sampled final spatial locations to within 7.75×10^{-9} at $T = 1$. An additional high-energy run at $(\rho_0, w) = (10, -0.5)$ and $T = 1.5$ agrees to 2.11×10^{-9}. Both codes integrate the same specified mathematical model and share its physical assumptions; this is an independent algorithmic cross-check, not an independent gravitational-field derivation.

A spatial convergence sequence at 81, 161 and 321 cells was also performed; the brane amplitude at $T = 1$ changes from -0.01144546 to -0.01144168 to -0.01144200 for the specified low-energy pulse. The data support numerical refinement of a smooth quantity, but are **not sufficient** to prove convergence of a discontinuous/thresholded causal signal front.

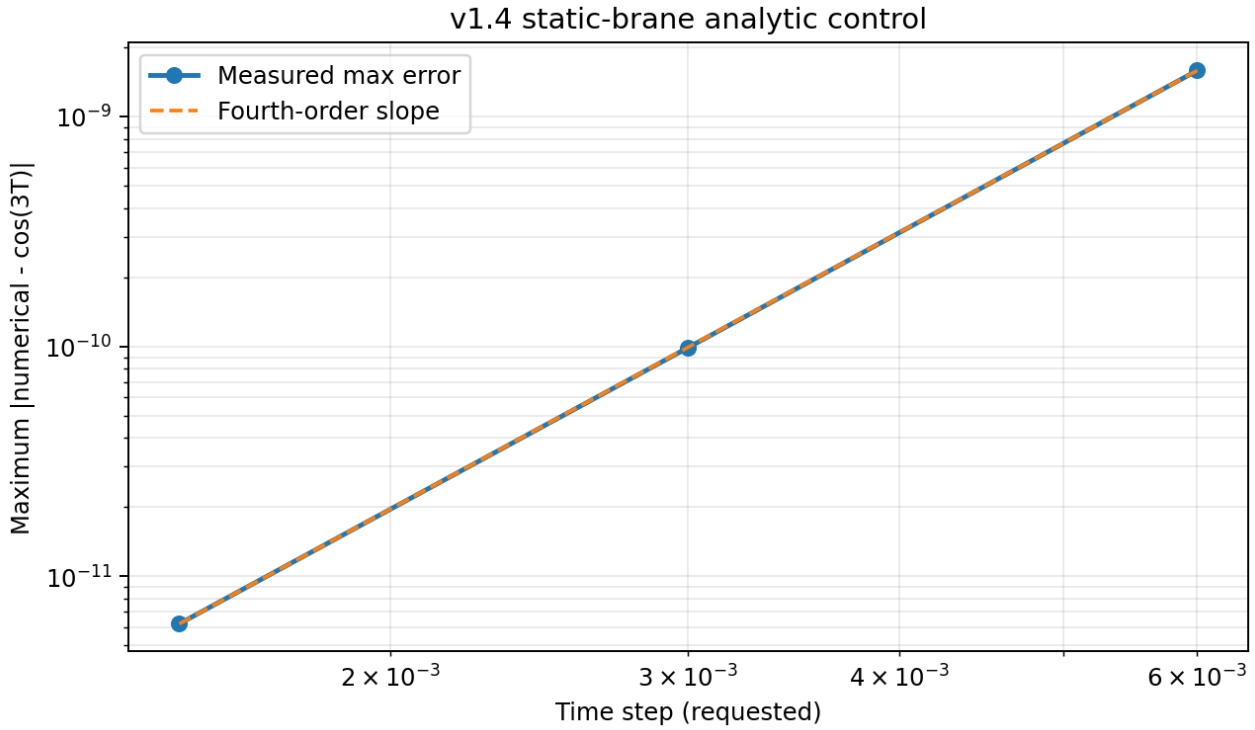


Figure 4: Static-brane RK4 convergence

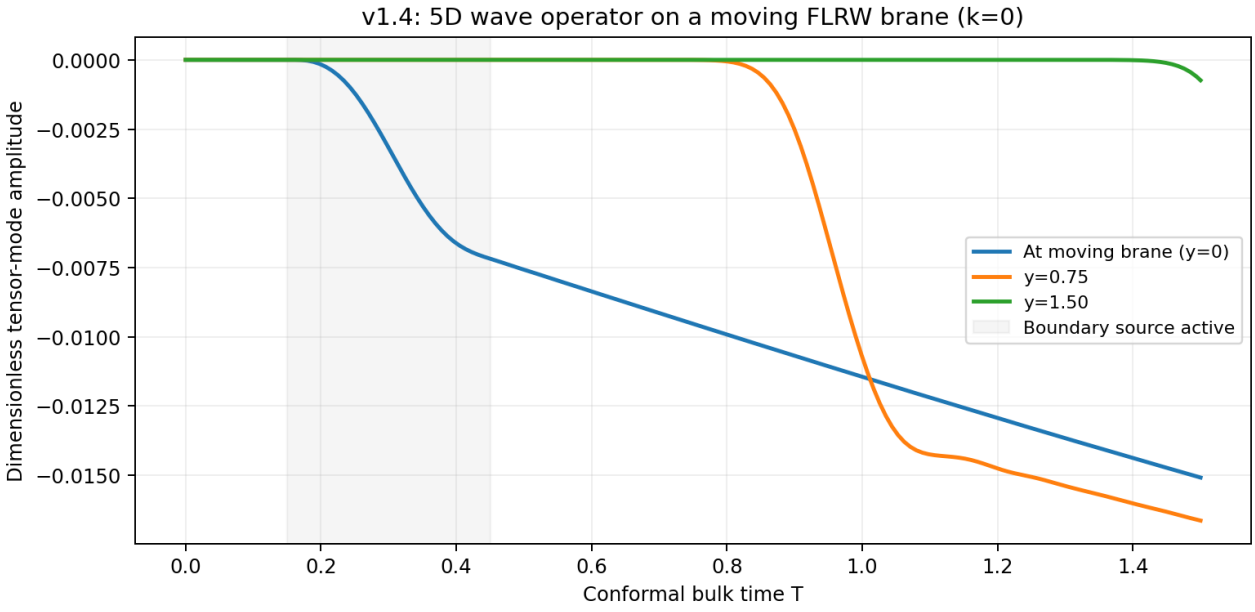


Figure 5: Dynamic wave fields

5. Planar Sources and the Signal-Front Problem

The additional 2D solver applies (4) to a source with temporal support $T \in (0.02, 0.12)$ and brane-coordinate support $0 \leq x < 0.12$. The comoving receiver is placed at $x = 1$. The example uses $\rho_0 = 10, w = -0.5$, corresponding to the *same high-energy background family* as the 20.96% geometric benchmark, although its pulse source and emitted events are different.

Independent integration of geometric null characteristics, for emission at $T = 0.02$, gives the following arrival bounds:

Emission point	Bulk-null arrival T	Brane-null arrival T	Difference in conformal T
Source center $x = 0$	1.36799126615	1.44163715130	0.07364588515
Closest source edge $x = 0.12$	1.25843305238	1.31751327373	0.05908022135

These are predictions of the **ray geometry** and are not identical to the original *receiver proper-time* percentage. A finite spatial source also changes the baseline path length. The planar code follows the field after injection and records the receiver amplitude.

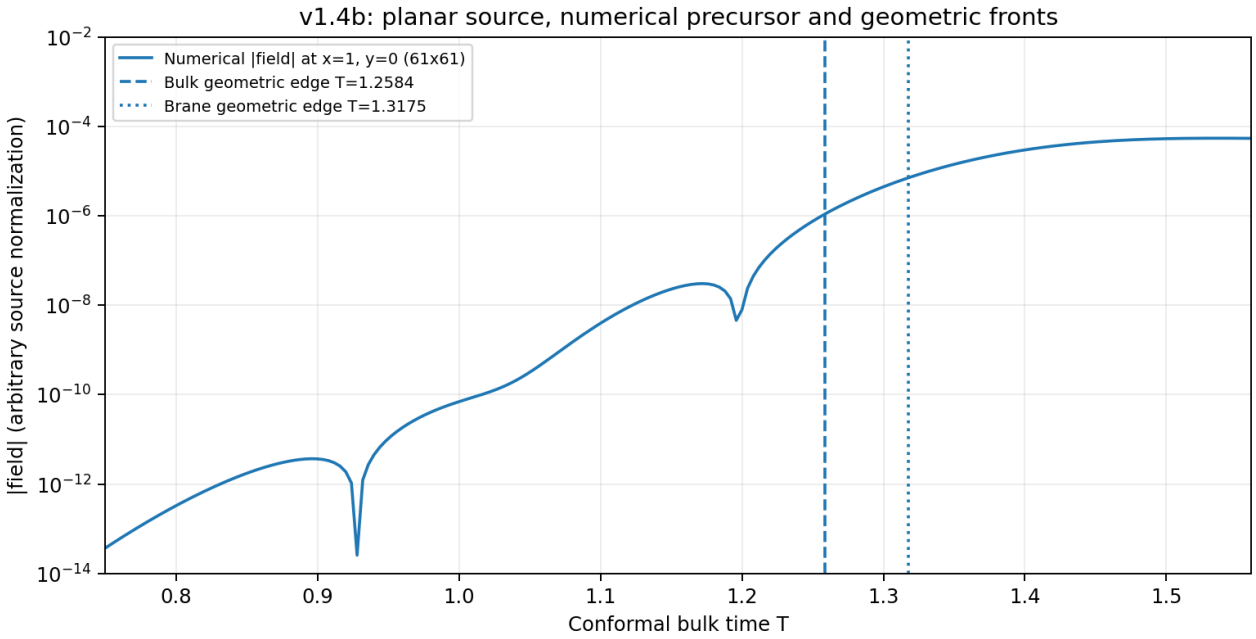


Figure 6: Pulse and geometric fronts

To test whether a numerical "first arrival" is meaningful, the field was sampled at two grids, 31x31 and 61x61, and five arbitrary absolute thresholds:

Absolute threshold $\lvert h \rvert$	First sampled T (31x31)	First sampled T (61x61)
10^{-12}	0.636	0.832
10^{-9}	0.848	1.076
10^{-7}	1.060	1.216
10^{-6}	1.176	1.260
10^{-5}	1.312	1.336

The huge variation is a **warning**, not evidence for superluminal field transmission. A semi-discretized hyperbolic equation can develop tiny nonzero numerical precursor amplitudes outside the continuum characteristic cone. An amplitude threshold selected after inspecting a result

is not an invariant causal arrival criterion. Without a continuum wavefront-convergence study and a calibrated physical source/detector, the 2D solver cannot establish an actual gravitational-wave shortcut time or observable amplitude.

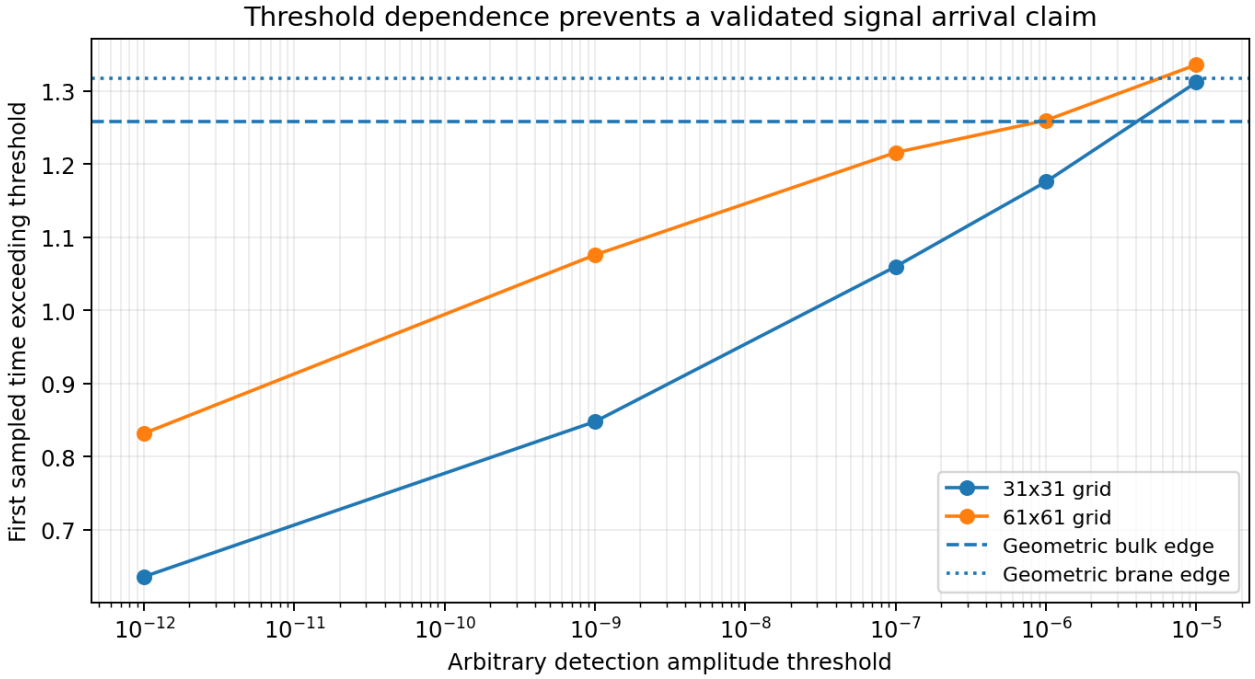


Figure 7: Threshold sensitivity

6. Geometric Travel-Time Advances in Physical Units

6.1 Parameters and cosmological model

To avoid incorrectly converting the dimensionless high-energy 20.96% result into seconds, the cosmological calculation constructs a **new low-energy scenario**. We use illustrative present-day reference numbers $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Omega_m = 0.315$, $\Omega_r = 9.2 \times 10^{-5}$ and $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$ [9]. For an AdS radius ℓ in megaparsecs, define $\bar{H}_0 = \ell H_0/c$. The tuned dimensionless total density today is evaluated in a numerically stable form,

$$\bar{\rho}_0 = \frac{6\bar{H}_0^2}{1+\sqrt{1+\bar{H}_0^2}}, \qquad \bar{\rho}(a) = \bar{\rho}_0\left(\Omega_m a^{-3} + \Omega_r a^{-4} + \Omega_\Lambda\right),$$

$$\bar{H}(a) = \sqrt{\frac{\bar{\rho}(a)}{3} + \frac{\bar{\rho}(a)^2}{36}}. \tag{5}$$

The source is placed at a comoving radial separation **40 Mpc**, a simple representative scale motivated by the approximate luminosity distance of GW170817 [7]. **Luminosity distance and comoving distance are not exactly identical**; this approximation and all adopted cosmological numbers are declared rather than inferred from a likelihood. The model is a flat brane with effective matter, radiation and vacuum terms, no dark radiation, no 5D perturbative source solution and no externally imposed empirical prior on ℓ. A cosmological parameter fit has *not* been run.

6.2 Stable positive-definite delay integral

Along a brane photon ray parametrized by comoving distance x, the embedding obeys

$$\frac{dT}{dx} = Q(x) = \sqrt{1 + \bar{H}(x)^2}, \qquad \frac{dz}{dx} = -\bar{H}(x).$$

For a travel interval of dimensionless comoving length D, write

$$T_* = \int_0^D Q(x)\,dx, \quad Z_* = \int_0^D \bar{H}(x)\,dx.$$

The bulk-null horizon at that same brane endpoint is $d_g = \sqrt{T_*^2 - Z_*^2}$, compared with the brane-photon distance D. Directly subtracting $d_g - D$ loses substantial relative precision when $\bar{H} \ll 1$. Introduce $\theta = \operatorname{arsinh} \bar{H}$ and the exact positive pair kernel

$$K(H_1, H_2) = \cosh(\theta_1 - \theta_2) - 1 = \frac{(H_1\sqrt{1+H_2^2} - H_2\sqrt{1+H_1^2})^2}{\sqrt{1+H_1^2}\sqrt{1+H_2^2} - H_1H_2 + 1}. \tag{6}$$

Since $Q^2 - H^2 = 1$, we obtain the exact identity

$$S := T_*^2 - Z_*^2 - D^2 = \int_0^D \int_0^D K[H(x), H(y)]\,dx\,dy \geq 0. \tag{7}$$

Therefore the **geometric excess reach** is robustly evaluated as

$$\delta x = \frac{S}{\sqrt{D^2 + S} + D}. \tag{8}$$

This positive double integral is evaluated on a midpoint grid, using a **single reusable paceval graph for (6)** and large batched evaluations of all distinct pairs; the Friedmann function and simplified observational chi-square are separately prepared paceval graphs. The receiver proper-time advance is estimated using a first-order inverse-endpoint expansion, $\delta\tau \simeq \delta x\, a_{\text{recv}}/(dd_g/dx)$ with $a_{\text{recv}} = 1$, and converted to seconds by the factor ℓ/c. For the tiny advances studied this inverse-endpoint approximation is independently checked against the **exact receiver root** at 68 decimal digits in an external reference calculation.

6.3 Accuracy and convergence

Two high-precision reference scenarios are:

ℓ (Mpc)	Independent direct-root advance (s)	paceval N=2000 midpoint (s)	paceval N=1000/2000 Richardson (s)
0.100	1.595174048923579	1.595173650111682	1.595174048925464
0.535	45.657868896584525	45.65785748155472	45.657868896582784

Midpoint errors decrease by approximately a factor four each time N doubles (100, 250, 500, 1000, 2000). The **Richardson-extrapolated values** achieve much higher arithmetic agreement in these selected cases; the conservative numerical error estimate stored in the CSV is $|\Delta_N - \Delta_{N/2}|/3$, rather than the much smaller realized difference to the reference. Repeated integration and root-finding at 68 decimal places validate the chosen double-precision algorithm **numerically**, not its physical assumptions.

The plot below illustrates how the **purely geometrical** result depends on the chosen AdS curvature length. The GW170817 delay line is shown only as an order-of-magnitude reference; it cannot be used as a direct exclusion threshold without a consistent emission and gravitational-wave propagation model.

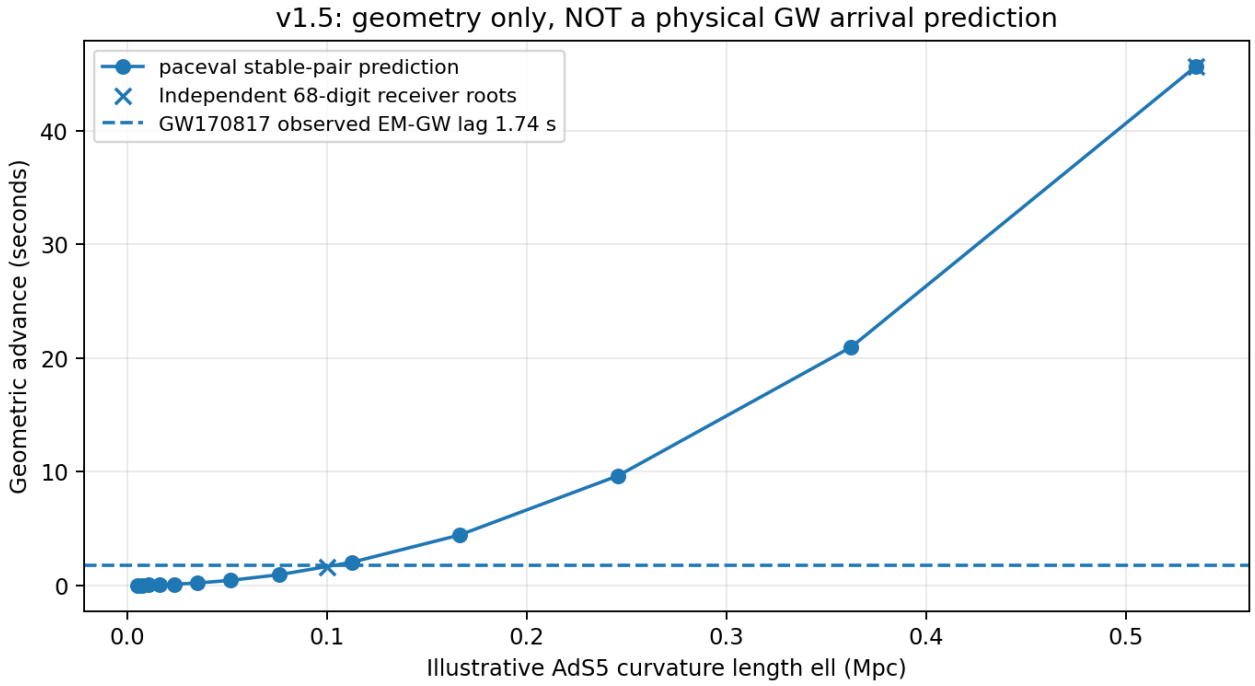


Figure 8: Illustrative geometric arrival-time advance versus the assumed AdS scale; not a prediction of the arrival time of a physical gravitational-wave signal.

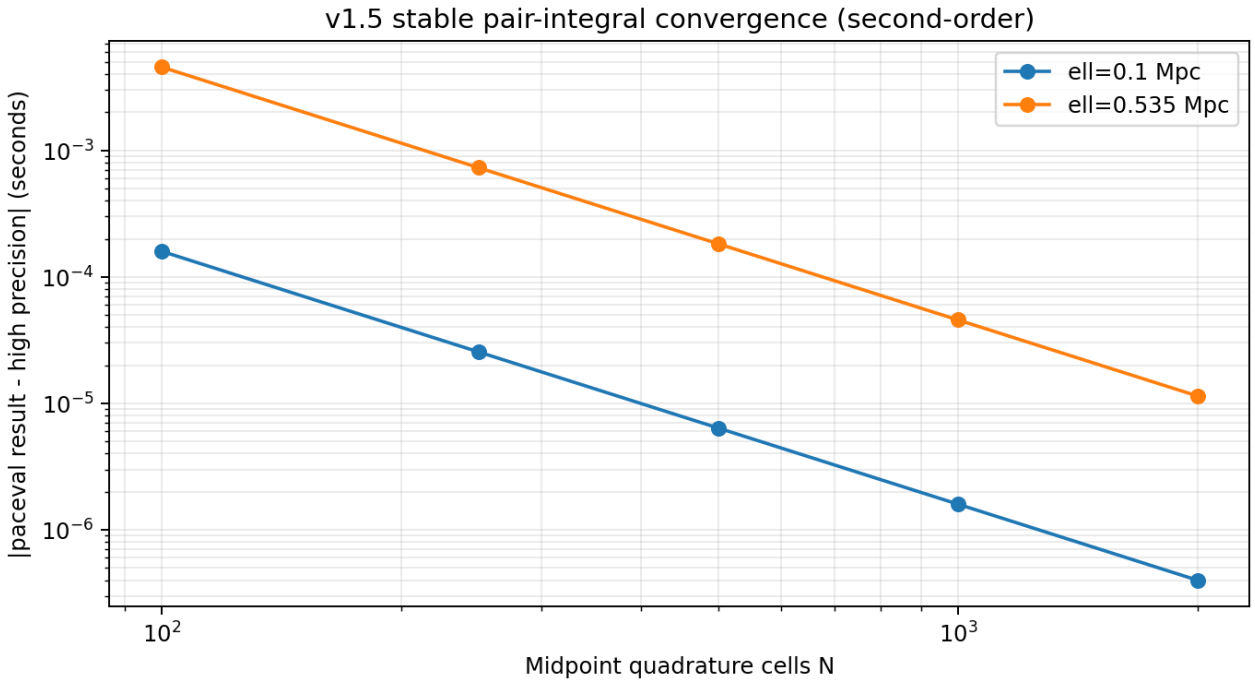


Figure 9: Numerical accuracy and convergence for the geometrical integral.

6.4 What GW170817 does and does not constrain in this experiment

The observed delay between GRB 170817A and GW170817 was (1.74 ± 0.05) seconds [7]. Introduce the unknown intrinsic emission-time difference t_{int} and an *assumed* geometric advance $t_{\text{geo}}(\ell)$. Then an illustrative timing equation is

$$\Delta t_{\text{obs}} = t_{\text{int}} + t_{\text{geo}}(\ell) + \epsilon. \tag{9}$$

The application evaluates $\chi^2 = [\Delta t_{\text{obs}} - t_{\text{int}} - t_{\text{geo}}(\ell)]^2/\sigma^2$ for three alternative nuisance assumptions: (i) $t_{\text{int}} = 0$; (ii) $t_{\text{int}} \in [0, 10]$ seconds, profiled; and (iii) t_{int} unrestricted, profiled. In case (iii), $t_{\text{int}} = \Delta t_{\text{obs}} - t_{\text{geo}}$ yields $\chi^2 = 0$ **for every parameter set**. In case (ii), the result depends on the hard lower limit and other model choices. These calculations are *not* Bayesian posteriors, confidence bounds, likelihood replicas or robust constraints on extra-dimensional gravity.

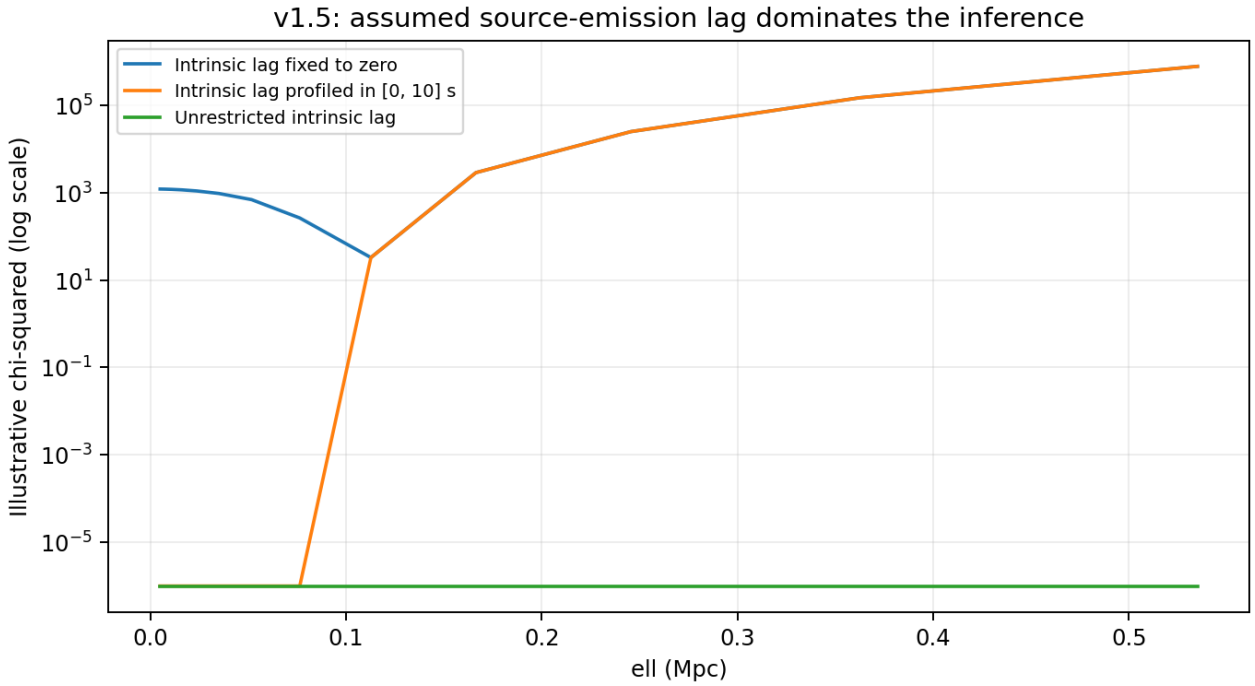


Figure 10: Degeneracy associated with the unknown intrinsic emission-time offset.

Visinelli, Bolis and Vagnozzi (2018) reported a model-dependent AdS5 curvature-length upper limit of approximately **0.535 Mpc at 68% confidence** from a significantly more complete likelihood analysis [8]. That number appears as a **bibliographic comparison scale**, not a limit obtained by our program. Our 45.66-second geometrical example at that scale does not invalidate their published result: the physical mapping, prior assumptions, emission-time treatment and likelihood construction must be aligned before results can be compared scientifically. Most importantly, the 5D wave solver in this package does not prove that the measured GW170817 signal would follow the geometric shortest route with a sufficiently observable amplitude.

7. Relation to Prior Literature and Originality

The wave-operator calculation follows established research on AdS_5 tensor perturbations. Easther et al. (2003), Hiramatsu, Koyama and Taruya (2005), and Seahra (2006) studied gravitational-wave evolution, higher-dimensional mode effects, high-energy cosmology, and methods for moving timelike brane boundaries [4–6]. The present source code is a reproducible implementation of a **restricted** wave problem. It does not establish a new gravitational-wave phenomenon or contradict known Kaluza–Klein mode leakage and amplitude-suppression results.

The physically scaled study is motivated by GW170817/GRB 170817A and existing braneworld constraints [7,8]. More recent models, such as Dai and Stojkovic (2024), use different bulk/brane

assumptions and do not independently confirm the particular parameters studied here [10]. The methodological contribution lies in the paceval graph-based implementation, the numerically stable positive pair-kernel identity, and the independent algorithmic cross-checks—including the finding that arbitrary amplitude thresholds do **not** yet yield a converged physical signal front.

8. Reproducibility and Numerical Cross-Checks

The study employs three functional **C++ programs using paceval**: a Fourier-mode solver, a planar pulse solver, and a solver for cosmologically scaled geometric travel-time differences. Mathematical expressions are evaluated through reusable paceval graphs. The documented numerical experiments include analytic controls and nontrivial moving-brane and parameter cases.

Separate numerical reference calculations, **not included in the production C++ programs**, were used to challenge the results. In particular, the wave-field calculations were compared against a separately written differential-equation implementation, and the small geometric time advances against high-precision quadratures and receiver root-finding. The preceding sections give numerical tolerances and convergence results.

For the two-dimensional pulse problem, identifying a first arrival still depends materially on the grid and amplitude threshold. That limitation must be distinguished from the good numerical reproducibility of smooth field quantities.

9. Scientific Limits, Falsification Criteria, and Future Work

The present work supports the **existence of the defined numerical solutions of a restricted wave-operator model** and the numerical accuracy of the dimensional geometric calculation. It **does not** establish (i) an excitable astrophysical 5D graviton wave packet with normalized coupling; (ii) a detector response to a bulk-returning wave with a measured earlier arrival; (iii) a continuum-converged front of a spatially compact 3D source; (iv) an observational AdS length posterior; (v) the existence of an accessible fifth dimension; or (vi) a viable mechanism for photons, spacecraft or matter to move through the bulk.

The highest-value next tasks are to derive the complete linearized Einstein-Israel tensor source boundary equations (with gauge and brane bending), replace the planar mathematical source by a physical 3D compact source and retarded Green-function calculation, implement a reflection-free outer boundary, certify wavefront convergence independently of amplitude thresholds, reconstruct detector-relevant spectra and then perform a properly marginalised GW/GRB timing-and-amplitude inference. A result must be rejected as an observed or measurable shortcut if it depends on a non-normalizable bulk mode, a numerical precursor, an unverified anisotropic-stress coupling or a source time-lag assumption hidden from the inference.

Conclusion

The paceval-based C++ work combines a geometric receiver calculation, a restricted AdS_5 wave-operator evolution, a planar pulse experiment, and physical-unit calculations of **geometric** travel-time differences. The principal positive findings are the reproducible 20.9628% conditional geometric benchmark, independently cross-checked numerical wave evolution, and a stable method for resolving extremely small cosmological range differences.

The model does **not** demonstrate that standard photons can travel through the bulk or that a realistically sourced, detectable gravitational-wave signal reaches a receiver earlier. Numerical field thresholds and uncertain source-emission lags prevent that stronger interpretation at present. The investigation advances the reproducibility and numerical methodology of braneworld shortcut research; it does **not** establish a new physical shortcut mechanism or its technological feasibility.

References

1. H. Ishihara (2001). *Causality of the Brane Universe*. Physical Review Letters 86, 381–384. https://doi.org/10.1103/PhysRevLett.86.381.
2. R. R. Caldwell and D. Langlois (2001). *Shortcuts in the Fifth Dimension*. Physics Letters B 511, 129–135. https://doi.org/10.1016/S0370-2693(01)00631-1.
3. E. Abdalla and A. G. Casali (2002). *Shortcuts in a Dynamical Universe and the Horizon Problem*. https://arxiv.org/abs/hep-th/0208008.
4. R. Easther, D. Langlois, R. Maartens, and D. Wands (2003). *Evolution of gravitational waves in Randall–Sundrum cosmology*. JCAP 10, 014. https://doi.org/10.1088/1475-7516/2003/10/014.
5. T. Hiramatsu, K. Koyama, and A. Taruya (2005). *Evolution of gravitational waves in the high-energy regime of brane-world cosmology*. Physics Letters B 609, 133–142. https://doi.org/10.1016/j.physletb.2005.01.024.
6. S. S. Seahra (2006). *Gravitational waves and cosmological braneworlds: A characteristic evolution scheme*. Physical Review D 74, 044010. https://doi.org/10.1103/PhysRevD.74.044010.
7. B. P. Abbott et al. (2017). *Gravitational Waves and Gamma-rays from a Binary Neutron Star Merger: GW170817 and GRB 170817A*. Astrophysical Journal Letters 848, L13. https://doi.org/10.3847/2041-8213/aa920c.
8. L. Visinelli, N. Bolis, and S. Vagnozzi (2018). *Brane-world extra dimensions in light of GW170817*. Physical Review D 97, 064039. https://doi.org/10.1103/PhysRevD.97.064039.
9. Planck Collaboration (2020). *Planck 2018 results. VI. Cosmological parameters*. Astronomy & Astrophysics 641, A6. https://doi.org/10.1051/0004-6361/201833910.
10. D.-C. Dai and D. Stojkovic (2024). *Superluminal propagation along the brane in space with extra dimensions*. European Physical Journal C 84, 175. https://doi.org/10.1140/epjc/s10052-024-12535-w.
11. paceval research (2026). *Autonomous Search and Multi-Objective Assessment of Theoretical Wormhole Geometries*. Preliminary English research report, 2 November 2026; not independently peer-reviewed.

Software provenance. Research calculations are based on the supplied paceval numerical programs, historical geometric benchmark data, and external high-precision validation records. The present report has not undergone independent physical peer review.